



United Energy Distribution Tariff Report 2009

3 November 2008

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Appendix 1 - Proposed 2009 DUoS and TUoS Tariff Rates

Appendix 2 – Glossary of Terms

1 Background

United Energy Distribution Pty Ltd (UED) is one of five electricity distribution businesses operating under licence within the State of Victoria, with assets totalling approximately \$2.0 billion. UED's network provides services to some 600,000 end-use customers in Melbourne's southern and eastern suburbs, with its area of operation confined to geographically defined boundaries set out in the Distribution Licence.

Distribution Use of System (DUoS) tariffs are designed to cover the cost of distribution of electricity to customers. However distributors bill retailers the Network Use of System (NUoS) charges.

NUoS tariff charges are comprised of two constituent parts, namely; Transmission Use of System (TUoS) tariffs and DUoS tariffs. The structures of the TUoS and DUoS tariffs are identical and the NUoS rates are the simple addition of DUoS tariffs and TUoS tariffs.

NUoS tariffs vary in structure depending on the voltage and size of the connection.

For practical reasons, UED's TUoS tariff structure does not reflect the structure of the grid fees paid by UED to Victorian Energy Networks Corporation (VENcorp) and SP Ausnet. The TUoS tariff structure has instead been aligned with the DUoS tariff.

In allocating transmission costs, UED has restricted the application of TUoS rates to those components of the NUoS rates which best reflect the underlying Grid Fees (i.e. Peak Energy, Summer Demand Incentive Charge, and Rolling Demand). Therefore, recovery of transmission related charges is mainly through peak energy, Summer Demand Incentive Charge (SDIC), and rolling demand charges, with off-peak energy and fixed charges not attracting TUoS.

Transmission costs to UED have been the subject of significant variability, with UED being limited in its ability to recover these costs, due to Commission imposed constraints on TUoS increases. Continued treatment of transmission costs in this manner exposes UED to significant risk in terms of cash flow, and distorts TUoS pass-through pricing signals to the market.

2 Purpose

This report is intended to give guidance to stakeholders, setting out the issues, principles and rationale used by UED in setting its NUoS tariffs for 2009. In so doing this report will meet the requirements set down in the Electricity Distribution Price review (EDPR) 2006-10 and incremental requirements and interpretations as laid out in the Comparative Report on the Distributors' Reporting Requirements (September 2006).

UED welcomes comments from interested parties for consideration in the continual evolution of UED's tariff and price setting.

3 Tariff Setting Principles (Basis for Tariff Policies)

The design of the tariffs sought to incorporate the following policy principles:

- **Regulatory compliance.** Compliance with the regulatory requirements of the Essential Services Commission EDPR 2006-10. A key requirement by the Commission is that the network prices faced by a group of customers must lie between the avoidable cost of servicing that group (lower bound) and the stand-alone costs of servicing that class (upper bound). These regulatory requirements limit the extent to which tariffs can change, particularly any average price increases.
- **Customer choice.** To provide customers with meaningful choices of tariff options based upon behavioural response and shared benefits.
- **Market equity.** Having regard to previous price levels and hence relative changes. Further, pricing should apply to all retailers in a neutral manner and not impede the viability of full retail contestability.
- **Cost reflectivity.** Seek appropriate mechanisms for achieving inter-customer group equity in the allocation of distribution cost-recovery requirements. Pricing to recognise cost-reflectivity within the constraints imposed by regulatory requirements (i.e. that tariffs fall within economic upper and lower bounds) and ensuring efficient investment signalling and network usage.
- **Behavioural elasticity.** Seek to utilise rational consumer behavioural elasticity, both in terms of usage responses to pricing signals and tariff switching.
- **Practicality.** Seek to simplify mechanisms balancing against economic functionality, while having regard to minimising transaction and pricing administration cost.
- **Environmental.** Within the limitations of the scope and context of electricity distribution pricing, have regard to opportunities to improve utilisation efficiency and accommodate emerging energy technologies, particularly in respect of greenhouse gas emission reduction.

Understandably, not all the principles can be satisfied to their full extent, while fully complying with the Commission's requirements, particularly as some pull in opposing directions. UED has sought to satisfy these principles to the greatest extent possible whilst maintaining the given regulatory constraints.

Annual tariff and price changes are governed by the requirements laid out in the 2006-10 EDPR, Codes, etc. The Annual Tariff Report details how the allowed increase in tariff basket prices have been applied, and also how the rebalancing constraints have been used. UED intends to gain the full revenue allowable by the tariff basket price controls to recover the efficient costs of operating the network business, including a commercial return on invested capital for "business as usual" service levels. UED also intends to use inter-tariff rebalancing where necessary to redress imbalances and cross subsidies. Tariff rebalancing has been used in the 2009 prices to lend greater emphasis to price signals as required by the tariff setting principles.

4 Changes in Tariffs, Tariff Structure and New Tariffs

4.1 Network Use of System Tariffs:

The 2009 NUoS tariffs reflect the underlying structure of both the TUoS and DUoS charges. That is, the structures of the TUoS and DUoS tariffs are identical and the NUoS rates are the simple addition of DUoS and TUoS.

The following tariffs are closed to new connections and only available to installations currently taking supply under these tariffs:

1. Low voltage kW time of use (LVkWTOU)
2. Low voltage large 1 rate (LVL1R)
3. Low voltage large 2 rate (LVL2R)
4. Low voltage medium 2 rate 5 day (LVM25D)
5. Low voltage medium 2 rate 7 day (LVM27D)
6. Low voltage small 2 rate (LVS2R)
7. Subtransmission KVA time of use (ST22kVATOU)

The tariffs that are open to new connections are as follows:

1. Dedicated circuit (Ded)

Please note that this tariff is not available to new or existing customers that install embedded generation. UED considers this a change in load characteristic and as such the tariff is not supported. The metering and data recording of such sites has additional regulated requirements to a standard site. It is technically not feasible to meet these requirements and at the same time be able to separately measure, control and bill a load controlled heating.
2. High voltage KVA time of use (HVkVATOU)
3. High voltage KVA time of use – HOT (HVkVATOUH)
4. Low voltage large KVA time of use (LVkVATOU)
4. Low voltage large KVA time of use – HOT (LVkVATOUH)
6. Low voltage kW time of use – HOT (LVkWTOUH)
7. Low voltage medium 1 rate (LVM1R)
8. Reverse cycle air-conditioning time of use (RCACKWTOU)
9. Low voltage small 1 rate (LVS1R)
10. Winter economy tariff (WET)
11. Unmetered supplies (UnMet)

Further information on each of the above tariffs and tariff eligibility is available on UED's website: www.ue.com.au.

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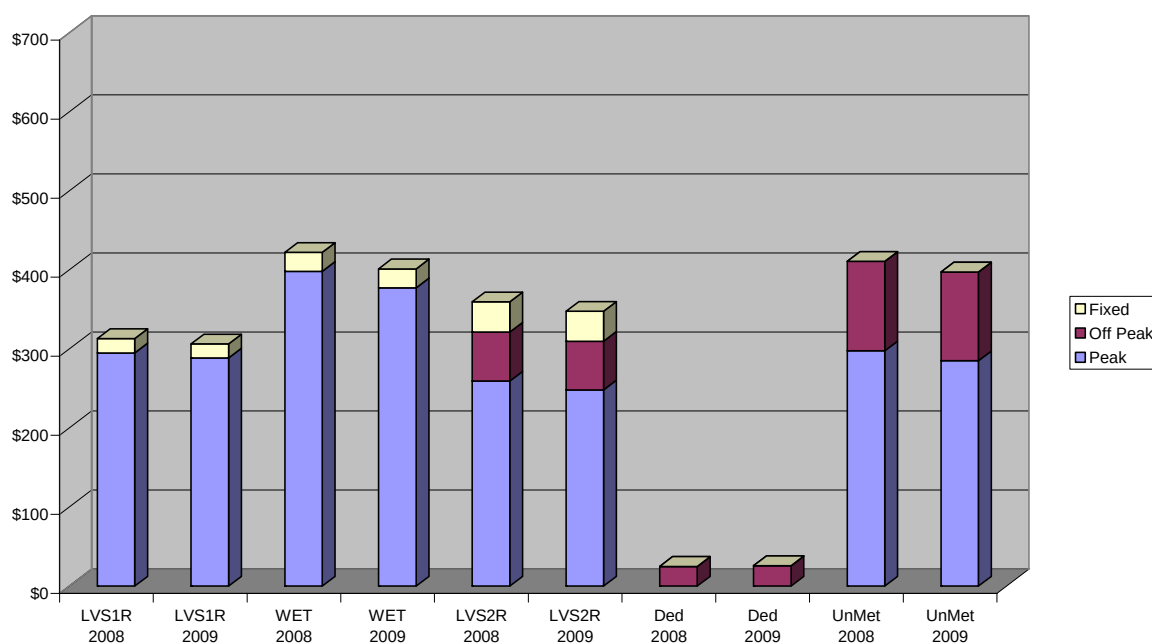
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UED believes it is important to provide customers with reasonable information on the objectives and rationale for tariff design and annual changes in tariffs. The Commission stipulates that this information must be provided in the form of a Tariff Strategy Report and updated by Annual Tariff Reports prior to any tariff changes.

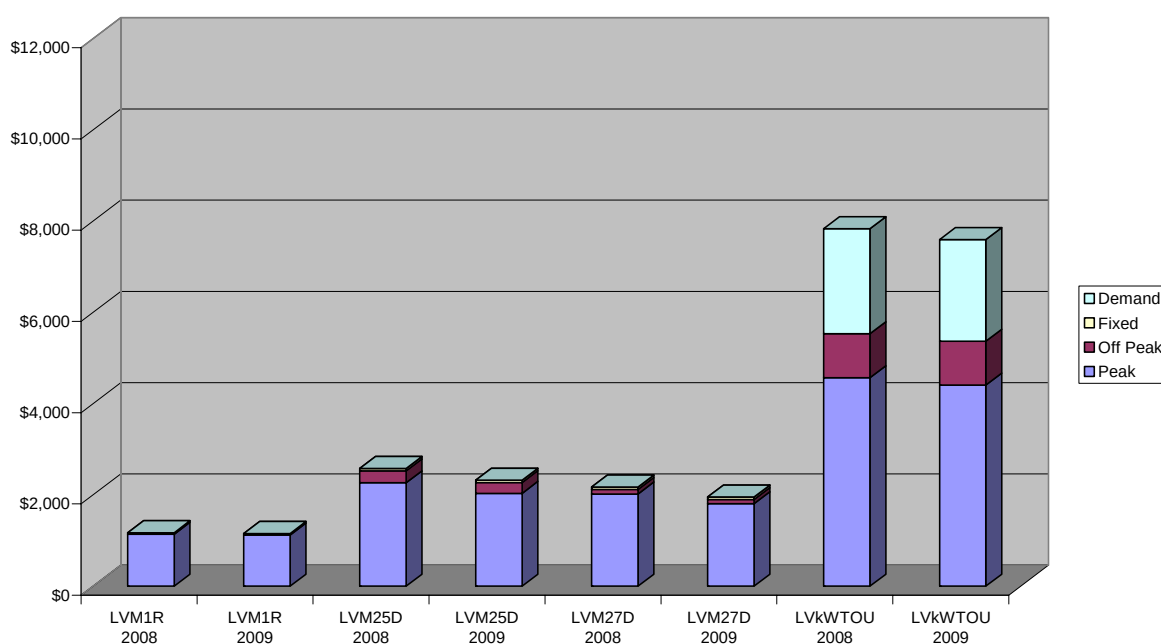
4.1.1 Average Charge per customer

This section presents the average yearly charge that UED's customers received in 2008 and estimated to receive in 2009. The following graphs are set out in the four main customer categories; small, medium, large and HV.

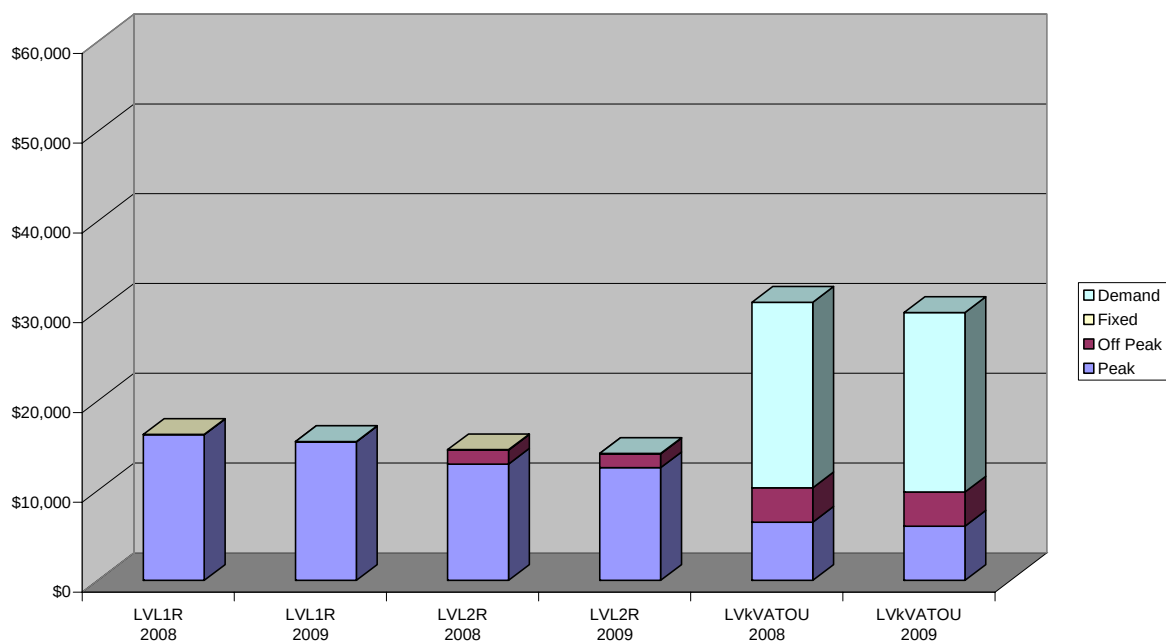
Average Distribution and Transmission charge per customer - Small



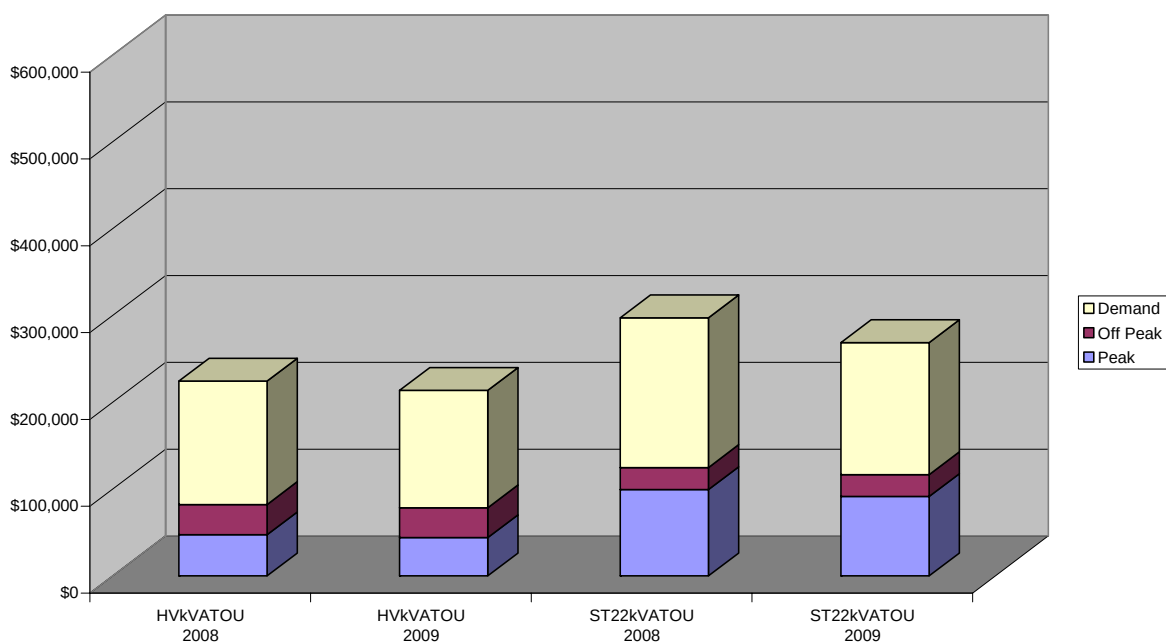
Average Distribution and Transmission charge per customer - Medium



Average Distribution and Transmission charge per customer - Large



Average Distribution and Transmission charge per customer - HV



4.2 Proposed Prescribed Services Tariffs

Under the 2006-10 EDPR, UED is allowed to recover its metering costs through Prescribed Metering Service Tariffs.

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Prescribed Services 2009 prices have been increased in accordance with the price control formulae by 25.98%. There has been no rebalancing undertaken, despite this being allowed in the price controls.

The Proposed Prescribed Metering Services and charges for 2009 are:

**Table 4.1 – 2009 Proposed Prescribed Metering Services Charges
(GST Exclusive)**

Metering Data Services	\$/meter/annum
Monthly read meter	\$65.610
Quarterly read meter	\$19.925
Metering Data Services (Unmetered Supplies)	\$/light/annum
Per light	\$1.099
Meter Provision Charges	\$/meter/annum
Single phase non off peak meter (single element)	\$2.391
Single phase off peak meter (two element)	\$6.296
Three phase direct connected meter	\$16.883
Three phase current transformer connected meter	\$26.556

4.3 Excluded Services

No changes to excluded services are proposed at this stage (except public lighting which is addressed in the following section). A list of these services and charges can be found on the UED web site: www.ue.com.au.

4.4 Public Lighting Excluded Service Charges

In accordance with the 2004 Review of Public Lighting Excluded Service Charges Final Decision of August 2004, UED has added revised Public Lighting Excluded Service Charges in the 2009 tariff approvals for application on 1 January 2009.

New Public Lighting charges have been submitted for approval for the following light types:

Table 4.2 - 2009 Proposed Public Lighting Charges (GST Exclusive)

Light Type	Excluded Service Charge \$ p.a. (Exclude G.S.T)
Fluorescent 2x20 watt	\$ 44.34
Fluorescent 3x20 watt	\$ 44.34
Mercury Vapour 50 watt	\$ 50.87
Mercury Vapour 80 watt *	\$ 34.37
Mercury Vapour 125 watt	\$ 50.87
Mercury Vapour 250 watt	\$ 50.36
Mercury Vapour 400 watt	\$ 69.73
Mercury Vapour 700 watt	\$ 69.73
Sodium High Pressure 70 watt	\$ 75.27
Sodium High Pressure 100 watt	\$ 60.50
Sodium High Pressure 150 watt	\$ 55.00
Sodium High Pressure 250 watt	\$ 55.34
Sodium High Pressure 400 watt	\$ 69.73
Metal Halide 70 watt	\$ 74.25
Metal Halide 100 watt	\$ 74.25
Metal Halide 150 watt	\$ 74.25
Metal Halide 250 watt	\$ 74.71
Metal Halide 400 watt	\$ 74.71

* Where a public lighting customer requires a lighting pole to be powder coated, the incremental costs of the powder coating will be negotiated with the customer.

The proposed adjustments to the 2009 Public Lighting service charges reflect an increase in the Public Lighting (PL) asset base ie. 2007 Public Lighting capital spend has been added to the PL asset base and is recovered from 2009, which is reflective of the 2009 average tariff increase.

Additionally, UED is committed to the introduction of Sustainable Public Lighting. Over the past 12 months, UED has worked with Councils, the Department of Sustainability and Environment (DSE) and the ESC to enable the introduction of Sustainable Public Lighting in UED.

UED have submitted T5 (2*14W Fluorescent Sustainable light) charges to the ESC and these charges are currently pending approval.

4.4.1 Public Lighting DUOS Charges

DUOS charges for public lighting have been, and continue to be set in the “Unmetered supplies” tariff. This is consistent with the 2004 public lighting review and the EDPR 2006-10.

4.5 Tariff Reassignments

UED does not intend to carry out tariff reassignments in 2009. Any tariff reassignments occurring during 2009 will be as a result of customer requests.

Retailers initiated 10 customer tariff reassignments inclusive of the period 1 January 2008 to 30 October 2008. As the tariff structures are not changing in 2009, it is expected that the number of reassignments in 2009 will be similar to 2008.

4.5.1 Tariff Assignment for New Connections

UED assigns new customers (new connections) coming on line in UED's territory for the first time the appropriate default network tariff. Determination of the appropriate network tariff is based on the customer's connection, expected annual energy consumption and/or maximum demand and type of metering. Table 4.3 outlines the categories based upon voltage, energy usage, and demand.

Table 4.3 – Network Tariff Categories

Category	Maximum Demand (kVA)	Annual Energy Consumption (MWh)
Small	< 20 and	< 70
Medium	20 to 150 and/or	70 to 400
Large	> 150 or	> 400

Table 4.4 below provides details of:

- Network tariff following contestability. These are the default tariffs that applied to former franchise retail tariffs following contestability. If required and with the necessary change of metering they can be changed to a corresponding 2009 eligible network tariff
- 2009 Eligible Network Tariffs (assuming the appropriate meter is installed). For new connections default tariffs apply. Upon submitting an application alternative tariffs listed are available provided an appropriate meter is installed. This applies to both new connections and existing customers

Note: Customers can apply to have their network tariff changed if they have had either a load characteristic change or a connection characteristic change. Determination of the appropriate network tariff for reassignment is based on the customer's expected annual energy consumption and/or maximum demand and type of metering. Customers may have to change their metering arrangement to be eligible for the revised tariff. To avoid gaming of

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tariffs, customers who choose a ToU tariff cannot move to another tariff for a minimum of 12 months.

With the continuation of the Dedicated tariff, new customers will be able to take the default LVS1R and LVM1R tariffs with a Dedicated circuit tariff which provides for all volume consumed between 11pm and 7am to be charged a lower unit rate.

**Table 4.4 – Network Tariffs following Contestability and 2009
Eligible Network Tariffs**

Former Franchise Retail Tariff	Network Tariff Following Contestability	Small/ Medium/ Large (based on annual energy and demand)	2009 Eligible Network Tariffs (appropriate metering required)
GD, GR	LVS1R LVkWTOU*	Small	LVS1R*** (Residential Customers Only) WET RCACLVkWTOU HOT LVkWTOU
		Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
GH, GL	LVS2R* LVkWTOU*	Small	LVS1R*** (Residential Customers Only) WET RCACLVkWTOU HOT LVkWTOU
		Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
E, G, N, B	LVM1R LVkWTOU*	Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
		Large	LVkVATOU HOT LVkVATOU
D	LVM2R5D* LVkWTOU*	Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
		Large	LVkVATOU HOT LVkVATOU
E1, B1, N1	LVM2R7D* LVkWTOU*	Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
		Large	LVkVATOU HOT LVkVATOU
L	LVL2R* LVkWTOU*	Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
		Large	LVkVATOU HOT LVkVATOU
V	LVL1R* LVkVATOU**	Medium	LVM1R*** (Business Customers Only) HOT LVkWTOU
		Large	LVkVATOU HOT LVkVATOU

* Indicates that these tariffs are only available to installations currently taking supply under this tariff. These tariffs are not available to new connections.

** Indicates that on becoming contestable these tariffs were available if an interval meter was installed.

*** Indicates the default network tariff for new connections.

4.6 Parent Tariffs

All of UED's tariffs can be provided on a standalone basis, and thus hierarchies of tariffs do not exist. UED defines Parent Tariffs as a predecessor to a current tariff - especially where the predecessor has been closed. As UED have not introduced any new tariffs from 2003 to 2009, there are no new parent tariffs.

4.7 Future Expenditure

Prices within the tariff structure are set in accordance with the Commission's 2006-10 EDPR which, from a regulatory perspective, accounts for the trade-off between future investment requirements and price/service levels. Also, UED's future expenditure as a result of network constraints can be found in UED's Transmission Connection Planning and Distribution (this is a bit out of date, can we get a more current report?) System Planning reports on the UED website: www.ue.com.au.

Planning reports:

One of the aims of the Planning Report is to provide information to would-be providers of non-network solutions (such as embedded generation or demand management) on emerging network constraints. UED's aim is to deliver '*least cost technically acceptable*' solutions to meeting demand, to the ultimate economic benefit of electricity consumers. To this end, proponents of non-network solutions to the emerging network constraints identified are invited to contact UED and lodge expressions of interest.

Embedded generation:

Sometimes a non-network solution to a constraint may take the form of embedded generation. UED hosts embedded generation as outlined in the Commission's *Electricity Industry guideline No.15 titled "Connection of Embedded Generation" dated August 2004*. That document also provides guidelines for the payment of network support fees to the generator where network savings result. The objective of the guideline is to provide clarity and transparency about arrangements for connecting embedded generating units to the distribution system, in particular in respect of:

- the way in which Distribution Network Service Providers (DNSPs) negotiate connection agreements with embedded generators;
- the charges under, and other terms and conditions of, connection agreements including principles DNSPs must observe in setting those charges and other terms and conditions;
- the payment to embedded generators of a share of a DNSPs avoided distribution system costs; and
- the payment to embedded generators of DNSPs avoided customer TUoS usage charges.

Demand management:

Another way of managing a network's ability to meet peak demand is to encourage customers to reduce load at times of the day and year when the demand is nearing the network's capacity to supply. Demand management initiatives come in a variety of forms including:

- 'behavioural' - Customers switching off discretionary loads when they know the network is approaching peak;
- 'controlled' - Customer making parts of their load available for the network Distribution operator to control remotely; and
- 'automated' - Installation of equipment that automatically limits peak demand.

Tariff pricing can be used to influence a customer's decision about whether or not to reduce load. If a short-term price signal is available then the customer may choose to reduce load during that price time zone. Alternatively, if some investment is required for additional equipment, then a customer could take into account the longer-term price signals.

At present, unfortunately technology often limits the possibilities. This may happen because:

- the customer does not know the network is congested (communications);
- it can be difficult to be sure that the peak demand has been reliably reduced every time it is needed and therefore a benefit has been produced (interval metering may assist); and
- few automatic devices are currently available.

In order for a real benefit to be produced, the reduction in peak must be reliable, so the network can count on the reduction and therefore perhaps deferring expenditure on additional capacity. If load is not curtailed, the constraint will still exist and UED can suffer further penalties in form of the Service Reliability Factor ("S" Factor).

UED has over the years created some innovative tariffs aimed at encouraging a demand-side response. Experience has shown however that a meaningful and material response is difficult to obtain. Essentially, most customers appear to value using their electricity more than the rewards available from curtailing or delaying consumption. It is also clear that (using simple metering) it is difficult to accurately measure and therefore reward peak reductions. Simple meters measure energy (kWh), but network peak costs are due to demand at a peak time (kW or kVA). UED remains hopeful however that developments such as the imminent widespread deployment of interval meters through AIMRO may enable the implementation of further, more effective pricing signals in future.

4.8 Contract demand charge resets

A customer can request a reduction in the maximum demand they get charged for if they are not currently signed onto a Supply Contract with a fixed Rolling Peak Minimum Demand charge.

For UED the base demand charge is calculated over a rolling 12-month period for Peak times only. From one month to the next, demand is re-calculated on the previous 12-month Peak period to the end of each bill. Therefore, an abnormal peak will be discarded out of the calculation after a 12-month period.

However, if a particular customer has changed their electricity usage over a shorter term (e.g. 3 - 6 months), then they are eligible (on their request) to accelerate their reduction in demand before the 12 month period is up.

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If a customer wants to reduce their chargeable demand, they must complete a "Demand Reset Form" off the UED website (www.ue.com.au). This form should be faxed or emailed to: UED Billing - 03 9256 5590.

From 1 January 2008 to 27 October 2008, 28 customers had their demand reset.

4.9 Apportionment to account for CPI estimates

In the Final Determination the ESC has allowed distributors to use actual CPI for the tariff setting process and therefore there is no requirement to account for the difference between actual and estimated CPI.

4.10 Comparisons with Tariff Strategy Report

UED has not introduced new tariff structures in 2009 and the rebalancing of tariffs and new or redundant tariffs has been discussed within this report. The UED 2009 proposed distribution tariffs are consistent with pricing strategies contained within the current Tariff Strategy Report 2006 – 2010.

4.11 Consultation and Stakeholder Management

UED did not conduct any information sessions during 2008 as there are no material tariff changes in 2009.

Consultation on UED related tariff issues is undertaken in conjunction with the Stakeholder Relationship Managers in accordance with the Stakeholder Business Plan.

Stakeholders are identified by UED, and are assigned to stakeholder groupings including customers, associations, retailers, affiliates, community, etc.

Over and above stakeholders identified by UED, stakeholders who wish to be included in community consultation are able to identify themselves to UED through the UED Service Desk on 1300 131 689, or emailing via talk-to@ue.com.au. The ability to communicate with UED via these channels is identified to potential participants on the UED website.

4.12 Details of Quantities and Customer Numbers Associated with Tariffs (Historical and Forecast).

**Table 4.5 – 2007 Actual, 2008 Estimate and 2009
Forecast Customer Numbers and Quantities**

Tariff	2007 Actual		2008 Estimate		2009 Forecast	
	Customers	MWhs	Customers	MWhs	Customers	Mwhs
Low voltage small 1 rate	546,269	2,606,258,505	549,848	2,724,630,992	553,168	2,710,758,674
Winter economy tariff	170	1,554,894	169	1,483,487	167	1,494,339
Reverse cycle air-conditioning time of use	-	-	-	-	-	1
Low voltage small 2 rate	5,139	39,022,462	5,117	39,969,473	5,057	39,281,157
Dedicated circuit	113,007	240,408,135	112,698	239,818,953	113,007	233,973,244
Low voltage medium 1 rate	30,117	420,175,369	31,329	450,334,519	32,153	446,930,428
Low voltage medium 2 rate 5 day	16,416	767,175,701	16,207	759,385,343	15,885	743,054,933
Low voltage medium 2 rate 7 day	6,793	219,822,440	6,749	219,660,938	6,659	206,546,637
Unmetered supplies	8,936	102,056,305	8,911	107,435,309	8,936	105,560,842
Low voltage large 1 rate	13	3,929,615	12	3,506,967	12	3,554,560
Low voltage large 2 rate	94	28,366,884	91	27,799,239	92	28,060,252
Low voltage KW time of use	852	128,234,766	841	127,283,491	841	128,476,269
Low voltage KW time of use - HOT	2	271,931	4	473,385	4	473,385
Low voltage large KVA time of use	2,341	2,353,754,222	2,582	2,349,377,373	2,608	2,470,258,957
Low voltage large KVA time of use - HOT	2	2,726,017	2	3,026,454	2	3,026,454

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Tariff	2007 Actual		2008 Estimate		2009 Forecast	
	Customers	MWhs	Customers	MWhs	Customers	Mwbs
High voltage KVA time of use	75	882,079,009	75	883,297,131	76	902,190,877
High voltage KVA time of use - HOT	1	62,759,795	1	74,067,879	1	74,067,879
Subtransmission KVA time of use	1	32,073,366	1	27,835,643	1	33,551,725
Total	730,227	7,890,669,414	734,636	8,039,386,575	738,668	8,131,260,612

5 Proposed Distribution Use of System (DUoS) Tariffs

Table 5.1 – 2009 Regulated Price Control Formulae

Component	% Increase/Decrease
DUOS	-0.84%
CPI	4.98%
L	-0.07%
X	2.50%
S	-3.05%

The Price control formula is:

$$\text{DUOS \% price movement} = (1+\text{CPI}) \times (1-\text{X factor}) \times (1+\text{S factor}) \times (1+\text{L factor})$$

The L-factor relates to Licence Fees paid by UED in the past financial year. The X-factor is the underlying price path for distribution tariffs over the regulatory period (2006-10). The S-factor relates to network reliability and reflects the network reliability statistics achieved by UED's management of its network assets.

Within the average tariff movement, the price control formulae allow individual tariffs to increase by a maximum of 3.74%.

The Commission has stated that distribution tariffs should lie between the following upper and lower bounds:

- tariffs for each customer should generate revenue in excess of the avoidable cost to service the customer; and
- tariffs for each customer should generate revenue less than the cost of providing the service on a stand-alone basis to the customer.

To demonstrate that distribution tariffs fall between the avoidable cost "floor" and standalone cost "ceiling", UED must first apply a "cost of supply" methodology to assist in setting tariff rates. Broadly speaking, tariff rates are set to recover the allocated distribution revenue from that customer group. It is noted, however, that UED has regard to all the pricing principles outlined in Section 3, in setting tariff rates.

The critical issue from a cost of supply modelling perspective is the method by which distribution revenue is allocated across the tariff groups. As network businesses are characterised by relatively high fixed costs and significant asset-sharing between customer groups, there is no unambiguously "correct" method for allocating costs. UED's method of allocation is based on each tariff's relative usage of UED's network assets.

In the model, customers are assigned into tariff groups based on voltage and demand characteristics. The consumption and demand characteristics for each tariff group are calculated.

- For asset based costs, the quantity of assets and supporting infrastructure are assigned to the tariff groups according to the combined consumption and demand characteristics of all customers using the asset, e.g. HV assets are assigned to LV and HV customers, but not to sub-transmission customers. The cost of providing the assigned assets is then calculated for each customer class.

- For operational and maintenance costs, costs are broken down into those that can be attributed to a particular asset class and the remainder are assigned to overheads
 - o Attributable costs use a weighted averaging to apply to the customers in each class
 - o Overheads are averaged over all customers

Combining the overhead, maintenance and infrastructure costs, the overall cost of supply for each customer is calculated.

UED has extended its “cost of supply” methodology to assess the avoidable and standalone costs. The avoidable cost model recognises that only a proportion of total costs are avoidable. In particular, the majority of asset-related costs cannot be avoided even if a particular customer group is no longer served. Inevitably, the assessment of which costs are avoidable is a matter of judgement. It should be noted, however, that as the avoidable costs are less than the total costs, UED’s cost of supply methodology will always set tariffs at a level that exceeds avoidable costs.

UED’s modelling of standalone costs is similarly based on the cost of supply model. The principal differences between the “basic” cost of supply estimates and standalone costs are:

- Standalone networks to serve a particular tariff class will not enjoy the benefit of diversity in peak demand between tariff classes;
- Economies of scale may be lost in supplying a subset of existing customers or tariffs;
- Greater urban congestion may result in the optimised replacement cost exceeding UED’s regulated asset value; and
- It is likely that a notional “standalone” competitor to UED may seek a rate of return that exceeds the regulated cost of capital.

These factors indicate that the standalone costs will exceed the cost of supply estimates on which UED bases its tariff design. It is important to recognise that it is difficult to determine the standalone costs with precision – inevitably a judgement must be made. The results of UED’s modelling is summarised in Table 5.2 below:

Table 5.2 - Comparison of 2009 Proposed Tariff Rates with Existing Estimated “Cost Window

Tariff Group		Lower Bound “Avoidable Cost” (c/kWh)	2009 Avg DUoS (Exc GST) (c/kWh)	Upper Bound “Standalone Cost” (c/kWh)
UnMet	Small	0.16	2.77	6.67
LVS1R			5.04	
LVS2R*			3.75	
Ded			1.21	
WET			3.29	
LVkWTOU*			4.17	
LVM1R	Medium	0.37	6.71	11.91
LVM2R5D*			4.34	
LVM2R7D*			4.99	
LVL2R*	Large	0.37	3.95	11.91
LVL1R*			3.72	
LVkVATOU	HV	0.14	2.43	4.36
HVkVATOU		0.07	1.25	2.14
ST22kVATOU*			0.37	

* Tariff closed to new connections and customers not already taking supply under this tariff.

5.1 Description of proposed price changes

In accord with the EDPR 2006-10 Price Determination, rebalancing has been undertaken of tariffs at the tariff component level.

This rebalancing takes into consideration and is consistent with the Price Determination and tariff policies, balancing the need to:

- recover maximum allowable revenue to recover the efficient costs of operating the network business,
- reduce risk in recovering revenue,
- give pricing signals to customers to provide an incentive for efficient utilisation of the network,
- be consistent with Pricing Principles and Cost of Supply Model where each tariff is:
 - above the avoidable cost of serving distribution customers,
 - below the cost of providing the service on a stand alone basis,
 - signal the impact of additional usage on future investment costs,

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- recover NUoS from customers in proportion to the services provided - classified by voltage, demand, and consumption patterns,
- be consistent with UED's tariff strategies,
- be consistent with the UED tariff policy framework.

Given the above considerations, it has been decided not to implement the Consumer Price Index Price Path (CPI-X) changes through a general across the board decrease in tariff rates as this would be inconsistent with the pricing principles which require signalling of the impact of additional usage on future investment costs. Accordingly some rebalancing has been undertaken at the tariff and tariff component level. A revised cost of supply model and other optimisation tools have been used to derive the final prices. Over and above the considerations listed above, the following provides details on how various types of tariffs have been affected by the rebalancing exercise and provide some general guidance by which to interpret the price changes.

The following general comments on intra-tariff rebalancing have been provided to indicate the flow on effects of the above tariff and price setting principles. While most tariffs are generally consistent with these comments, individual tariffs may vary slightly due to the overall optimisation process.

Intra-tariff rebalancing general comments look at the tariff components (such as fixed charges, and peak energy charges) and how they have been affected by the 2009 price proposals:

- Summer Demand Charges have increased slightly as this is the most targeted pricing signal for network optimisation.
- Peak energy charges have generally been slightly increased to discourage usage in peak times.

Table 5.3 – 2009 Estimated Percentage change in Proposed DUoS Tariffs

Tariff	Movement
Low voltage small 1 rate	1.2%
Winter economy tariff	-0.8%
Reverse cycle air-conditioning time of use	-0.8%
Low voltage small 2 rate	-0.8%
Dedicated circuit	3.7%
Low voltage medium 1 rate	1.4%
Low voltage medium 2 rate 5 day	-9.2%
Low voltage medium 2 rate 7 day	-9.2%
Unmetered supplies	-0.8%
Low voltage large 1 rate	-0.8%
Low voltage large 2 rate	-0.8%
Low voltage KW time of use	-0.8%
Low voltage KW time of use - HOT	-0.8%
Low voltage large KVA time of use	0.1%
Low voltage large KVA time of use - HOT	-0.8%
High voltage KVA time of use	0.0%
High voltage KVA time of use - HOT	-0.8%
Subtransmission KVA time of use	-0.9%

6 Proposed TUoS tariffs

For practical reasons, the TUoS tariff structure does not reflect the structure of the grid fees paid. The TUoS tariff commenced with a structure compatible with the DUoS tariff. The structure has been maintained in order to allow the NUoS tariff to be determined by simply adding the DUoS and TUoS rates. However, UED has restricted the application of TUoS rates to those components of the NUoS which best reflect the underlying Grid Fees (i.e. Peak Energy, Summer Demand Incentive Charge and Rolling Demand). Therefore, off peak energy and fixed charges do not attract TUoS.

TUoS revenue receivable is governed by a Maximum Transmission Revenue equation as detailed in the EDPR. For 2009 this equation yields a TUoS decrease of 9.34%, which has been applied evenly across all Tariffs (see table 6.1 below).

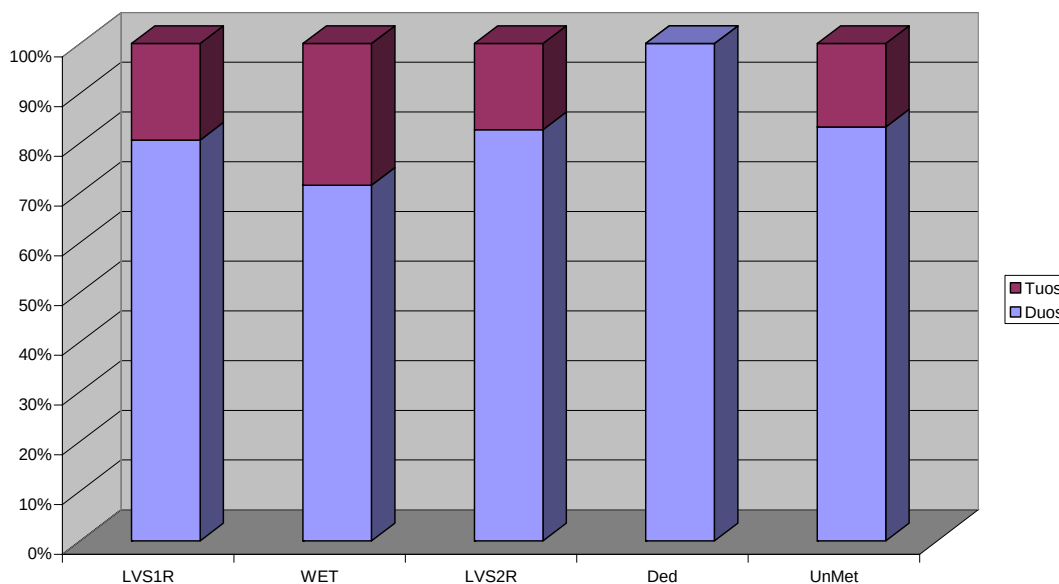
Table 6.1 – 2009 Estimated Percentage change in Proposed TUoS Tariffs

Tariff	Movement
Low voltage small 1 rate	-9.3%
Winter economy tariff	-9.4%
Reverse cycle air-conditioning time of use	-9.3%
Low voltage small 2 rate	-9.4%
Dedicated circuit	NA
Low voltage medium 1 rate	-9.3%
Low voltage medium 2 rate 5 day	-9.4%
Low voltage medium 2 rate 7 day	-9.3%
Unmetered supplies	-9.4%
Low voltage large 1 rate	-9.4%
Low voltage large 2 rate	-9.4%
Low voltage KW time of use	-9.4%
Low voltage KW time of use - HOT	-9.4%
Low voltage large KVA time of use	-9.3%
Low voltage large KVA time of use - HOT	-9.3%
High voltage KVA time of use	-9.3%
High voltage KVA time of use - HOT	-9.4%
Subtransmission KVA time of use	-9.3%

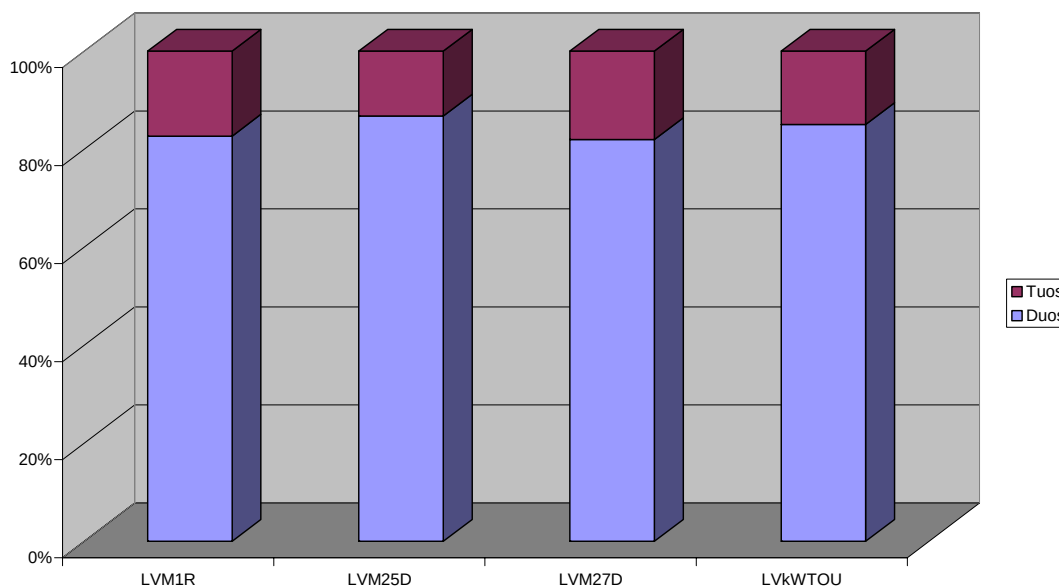
6.1 Average percentage split between TUoS and DUoS

This section presents the average percentage split between TUoS and DUoS that UED's customers will receive in 2009. The following graphs are set out in the four main customer categories; small, medium, large and HV.

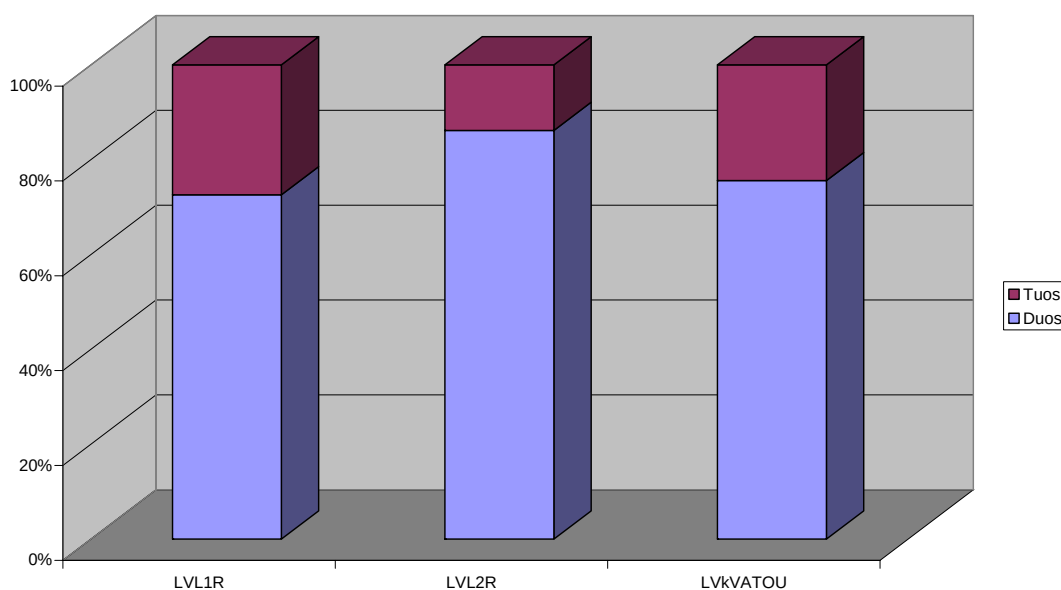
2009 DUOS / TUOS Estimated Revenue percentage of NUOS - Small



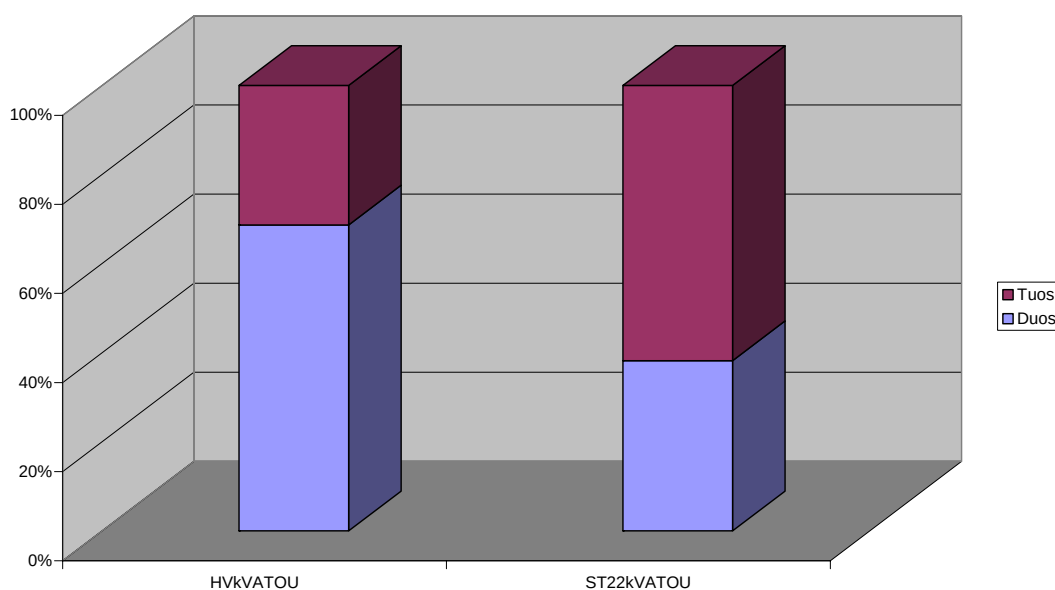
2009 DUOS / TUOS Estimated Revenue percentage of NUOS - Medium



2009 DUOS / TUOS Estimated Revenue percentage of NUOS - Large



2009 DUOS / TUOS Estimated Revenue percentage of NUOS - HV



6.2 Annual Adjustment

TUoS inter tariff rebalancing is allowed under the EDPR, with a Y factor setting the maximum rebalancing over and above a CPI increase over the previous years prices, but in totality remaining below the Maximum Transmission Revenue. For 2009, UED has proposed an overall decrease which applies across the board to all tariffs and all tariff components.

7 Proposed Network Use of System (NUoS) Tariffs

NUoS charges are those seen on user invoices and reflect the underlying structure of Transmission and Distribution charges. Residential and small business customers do not see distribution prices explicitly, as they are usually bundled by retailers with energy and

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other costs. NUoS rates are the simple addition of DUoS and TUoS rates. As a result of the movements discussed above, the proposed NUoS rates are as follows:

Table 7.1 – Schedule of Network Use of System (NUoS) Tariffs: 1 January 2009 (GST Exclusive)

Description	Tariff Code	Network Tariff Component							Eligibility (consumption/ category)		Minimum Chargeable Rolling Demand
		Standing Charge (c/day)	Summer Peak Energy (c/kWh)	Non Summer Peak Energy Block 1 (c/kWh)	Non Summer Peak Energy Block 2 (c/kWh)	Off Peak Energy (c/kWh)	Rolling Peak Demand c/kVA/day	Summer Demand Incentive Charge c/kW/day or c/kVA/day	kVA	MWh pa	kVA
Low voltage small 1 rate	LVS1R	4.892	7.335	4.981					Residential Customer		
Low voltage small 2 rate*	LVS2R*	10.329	9.685	7.442		1.286					
Dedicated circuit	LVDed					1.214					
Unmetered supplies	UnMet		9.497	7.094		1.330					
Winter Energy Tariff**	WET2Step**	6.513	6.371	4.810	2.559				<20	<70	
Reverse cycle airconditioning time of use	RCACKWTOU		14.475	3.344		1.042		59.534	<20	<70	
Low voltage medium 1 rate	LVM1R	9.538	9.988	6.511					Business Customer		
Low voltage medium 2 rate 5 day*	LVM2R5D*	13.694	7.846	6.398		1.355					
Low voltage medium 2 rate 7 day*	LVM2R7D*	14.707	7.976	6.673		1.272					
Low voltage KW time of use*	LVkWTOU*		7.045	4.362		1.382		33.602	<150	<400	
Low voltage KW time of use - HOT	LVkWTOUH		6.013	4.645		1.202		53.040	<150	<400	
Low voltage large 1 rate*	LVL1R*	9.198	5.797	4.590							
Low voltage large 2 rate*	LVL2R*	13.267	7.811	6.217		1.254					
Low voltage large KVA time of use	LVkVATOU		1.624	1.345		0.781	11.807	17.257	>150	>400	150
Low voltage large KVA time of use - HOT	LVkVATOUH		1.266	1.037		0.679	11.313	27.170	>150	>400	150
High voltage KVA time of use	HVkvATOU		1.048	0.886		0.486	8.955	11.249	>150	>400	1,150
High voltage KVA time of use - HOT	HVkvATOUH		0.870	0.713		0.483	8.576	20.019	>150	>400	1,150
Subtransmission KVA time of use*	SubTkVATOU		0.655	0.518		0.220	2.902	3.667	>150	>400	11,100
*Tariff closed to premises not already taking supply under this tariff and new connections **Winter Energy Tariff - Step 1 <=1020kwh/Quarter, Step 2 >1020kwh/Quarter Non-summer peak energy rates apply for period 1st April to 31st October											

8 Future Direction of UED's Tariffs

UED will continue to review the effectiveness of its existing tariffs. UED has been a leader in tariff innovation and will assess tariff design implications as more accurate half-hourly load shapes for all customers becomes available.

It is expected that while there may be some refinements to existing tariffs, UED is anticipating the introduction of a new tariff in 2010 based on 'time of use' to provide pricing signals following the commencement of the AIMRO meter roll-out. The design and pricing is still in development, however the principals detailed below will be adhered to.

UED will continue to maintain a watching brief on the need for more substantive tariff changes taking into account the following:

- Encouraging an increased uptake of interval-metering based tariffs such as the current 'ToU' series;
- The SDIC concept will remain, but the time window may be updated from time to time in order that it remain aligned with the key network peak demands;
- Cost-of-supply modelling updated to reflect changes in relative contributions from segments;
- Further accentuating the emphasis on peak season (summer), day of week and time of day in order to stimulate Demand Side Management (DSM) response;
- Properly integrate the contribution that distribution-connected generator customers should be making to the costs of providing network services that all users share and derive value from;
- Further closure of tariffs based on obsolete metering;
- Premium service tariffs whereby customers get a choice of above code-level supply reliability and services, for a premium on top of the standard tariff. This must be seen in the overall context of customer service as well as relationship strategies;
- An increased number of time-of-day bands, with greater peak / off peak differential, and energy and distribution tariff components peaking at different times; and
- Demand management (DM) programs aimed at different customer classes may be investigated, for example:
 - interruptible tariffs for business customers whereby customers agree to reduce their power consumption for agreed periods at the request of the distributor (likely to be at a time like a hot summer afternoon when the time when the system is heavily stressed), and in return get some compensation payments from the distributor; and
 - DM aggregation program, which involves working with a range of customers and bidding their combined interruptible load in either the wholesale energy or ancillary services market.
- Investigate positive pricing incentives such as rewards and rebates as motivational mechanisms for DM.

8.1 Premium Feed in Tariffs

The Victorian Government is considering a policy for Premium Feed in Tariffs in Victoria. The policy is yet to be passed in legislation and may be subject to further debate or change. UED is an active participant in ensuring policy implementation. A brief overview of the policy is:

- Premium Feed in Tariffs (PFIT) will apply only to residential customers with photovoltaic (PV) cells up to 2kW.
- The minimum PFIT is 60 cents / kWh (inc GST) applied to the net generation
- The Distribution Business provides the minimum PFIT (or higher tariff) to the Retail Business who charges the customer
- The policy will apply for 15 years and is targeted to commence 1 January 2009 (although this may be delayed)
- The policy will apply to the first 100 MW of installed PV capacity over 15 years across Victoria

Policy Implementation:

- The Government will introduce a Bill to parliament which will give effect to this policy from November 2008
- The Bill will provide the Distribution Businesses a legislative pass through in the current EDPR period for the costs ie. it will define the policy as a relevant pass through event in the EDPR Part 2.

Appendix 1 - Proposed 2009 DUoS and TUoS Tariff Rates

Table A.1 – Schedule of Distribution Use of System (DUoS) Tariffs:

1 January 2009 (GST Exclusive)

Description	Tariff Code	Network Tariff Component							Eligibility (consumption/ category)		Minimum Chargeable Rolling Demand
		Standing Charge (c/day)	Summer Peak Energy (c/kWh)	Non Summer Peak Energy Block 1 (c/kWh)	Non Summer Peak Energy Block 2 (c/kWh)	Off Peak Energy (c/kWh)	Rolling Peak Demand c/kVA/day	Summer Demand Incentive Charge c/kW/day or c/kVA/day	kVA	MWh pa	kVA
Low voltage small 1 rate	LVS1R	4.892	5.983	3.851					Residential Customer		
Low voltage small 2 rate*	LVS2R*	10.329	7.414	5.623		1.286					
Dedicated circuit	LVDed					1.214					
Unmetered supplies	UnMet		7.369	5.392		1.330					
Winter Energy Tariff**	WET2Step**	6.513	4.857	3.599	1.348				<20	<70	
Reverse cycle airconditioning time of use	RCACKWTOU		4.220	1.061		1.042		53.032	<20	<70	
Low voltage medium 1 rate	LVM1R	9.538	8.384	5.230					Business Customer		
Low voltage medium 2 rate 5 day*	LVM2R5D*	13.694	6.650	5.441		1.355					
Low voltage medium 2 rate 7 day*	LVM2R7D*	14.707	6.386	5.400		1.272					
Low voltage KW time of use*	LVkWTOU*		5.937	3.475		1.382		28.432	<150	<400	
Low voltage KW time of use - HOT	LVkWTOUH		5.316	4.088		1.202		46.791	<150	<400	
Low voltage large 1 rate*	LVL1R*	9.198	4.212	3.323							
Low voltage large 2 rate*	LVL2R*	13.267	6.593	5.241		1.254					
Low voltage large KVA time of use	LVkVATOU		0.930	0.788		0.781	8.971	13.382	>150	>400	150
Low voltage large KVA time of use - HOT	LVkVATOUH		0.828	0.685		0.679	9.524	22.466	>150	>400	150
High voltage KVA time of use	HVkvATOU		0.560	0.495		0.486	5.708	7.840	>150	>400	1,150
High voltage KVA time of use - HOT	HVkvATOUH		0.590	0.489		0.483	6.715	16.261	>150	>400	1,150
Subtransmission KVA time of use*	SubTkVATOU		0.326	0.253		0.220	0.569	0.835	>150	>400	11,100
*Tariff closed to premises not already taking supply under this tariff and new connections **Winter Energy Tariff - Step 1 <=1020kwh/Quarter, Step 2 >1020kwh/Quarter Non-summer peak energy rates apply for period 1st April to 31st October											

Table A.2 – Schedule of Transmission Use of Systems (TUoS) Tariffs:
1 January 2009 (GST Exclusive)

Description	Tariff Code	Network Tariff Component							Eligibility (consumption/ category)		Minimum Chargeable Rolling Demand
		Standing Charge (c/day)	Summer Peak Energy (c/kWh)	Non Summer Peak Energy Block 1 (c/kWh)	Non Summer Peak Energy Block 2 (c/kWh)	Off Peak Energy (c/kWh)	Rolling Peak Demand c/kVA/day	Summer Demand Incentive Charge c/kW/day or c/kVA/day	kVA	MWh pa	kVA
Low voltage small 1 rate	LVS1R		1.352	1.130					Residential Customer		
Low voltage small 2 rate*	LVS2R*		2.271	1.819							
Dedicated circuit [#]	LVDed [#]										
Unmetered supplies	UnMet		2.128	1.702							
Winter Energy Tariff**	WET2Step**		1.514	1.211	1.211				<20	<70	
Reverse cycle airconditioning time of use	RCACKWTOU		10.255	2.283				6.502	<20	<70	
Low voltage medium 1 rate	LVM1R		1.604	1.281					Business Customer		
Low voltage medium 2 rate 5 day*	LVM2R5D*		1.196	0.957							
Low voltage medium 2 rate 7 day*	LVM2R7D*		1.590	1.273							
Low voltage KW time of use*	LVkWTOU*		1.108	0.887				5.170	<150	<400	
Low voltage KW time of use - HOT	LVkWTOUH		0.697	0.557				6.249	<150	<400	
Low voltage large 1 rate*	LVL1R*		1.585	1.267							
Low voltage large 2 rate*	LVL2R*		1.218	0.976							
Low voltage large KVA time of use	LVkVATOU		0.694	0.557			2.836	3.875	>150	>400	150
Low voltage large KVA time of use - HOT	LVkVATOUH		0.438	0.352			1.789	4.704	>150	>400	150
High voltage KVA time of use	HVkvATOU		0.488	0.391			3.247	3.409	>150	>400	1,150
High voltage KVA time of use - HOT	HVkvATOUH		0.280	0.224			1.861	3.758	>150	>400	1,150
Subtransmission KVA time of use*	SubTkVATOU		0.329	0.265			2.333	2.832	>150	>400	11,100
*Tariff closed to premises not already taking supply under this tariff and new connections											
**Winter Energy Tariff - Step 1 <=1020kwh/Quarter, Step 2 >1020kwh/Quarter											
[#] Dedicated Tariff - not applicable to TUOS											
Non-summer peak energy rates apply for period 1st April to 31st October											

Appendix 2 - Glossary of Terms

DSM	Demand-side management. Refers to the ability of customers that consume a product to change behaviours that alter the overall balance between supply and demand. Examples may include the installation of energy management systems, or deferring consumption at time of peak demand.
The Commission	Essential Service Commission.
LVS1R etc	Abbreviated Tariff names. • The 1st two letters refer to the voltage level at which the customer is supplied, LV = low voltage 240V (single phase) or 415V (three phase), HV = high voltage 6.6, 11 or 22kV, ST = sub-transmission 22 and 66kV. HOT = tariff variant based on high ambient temperature threshold. • The 2nd section refers to the type of customer, S = small, M = medium, L = large, no letters generally indicates 'generic' interval metered customer. • 3rd section indicates type of rate structure, 1R = single energy rate, 2R = peak/off-peak rate, kW = interval metered in kW, kVA = interval metered in kVA.
kW	Kilowatts, (1,000 Watts) the ISO unit of power measurement.
kWh	An I.S.O. unit used for the measurement of energy. It is the product of kW times time (in hours).
kVA	kilo-Volt-Ampere, the vector product of voltage times current. It is a measure of the proportion of useful (kW) power delivered vs. the current (Amps) drawn by the load, this is known as 'power factor'. The size of assets required is generally a function of the current. Therefore the closer the power factor is to 1.00, the smaller the asset required to supply the load.
Residue settlement	A means by which over-recoveries of costs on the part of the energy and transmission markets are rebated back to customers.
SDIC	Summer Demand Incentive Charge.
DPI	Department of Primary Industry
IMRO	Interval Meter Rollout
AIMRO	Advanced Interval Meter Rollout
OIC	Order in Council